Homework Assignment 3
Image Filtering
& Intensity Transformation

Digital Image Processing

Deadline: 09:00 AM, 11/10/2025

HW 3(A) Problem Statement

Adopt the steps outlined in Section 4.7 to repeat Example 4.15, pp. 271-273 using the vertical Sobel kernel shown in Fig. 4.38(a) and the test image “keyboard.tif” provided along with this homework assignment. You may use any existing library to compute the Fourier transform.

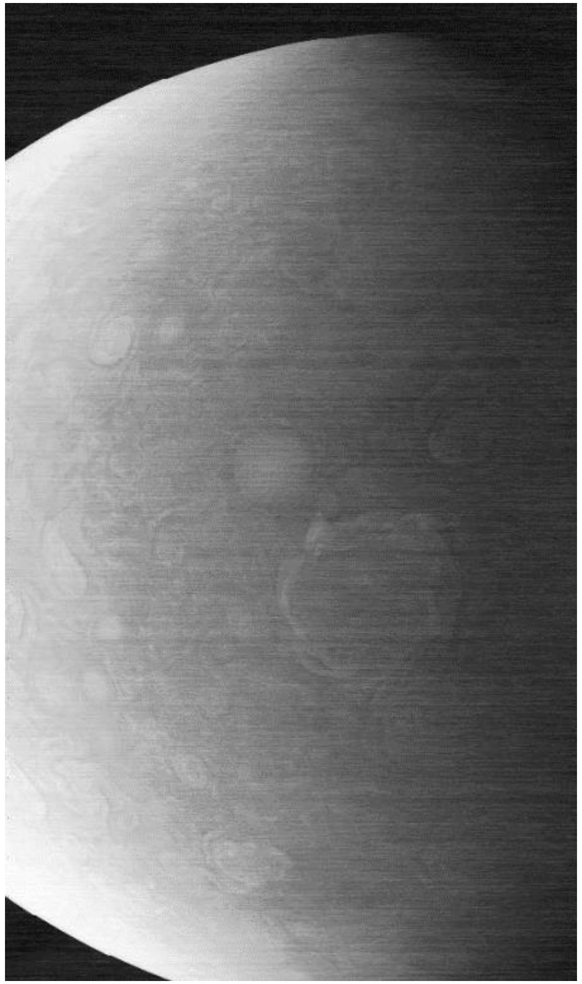
1) (5%) Show the Fourier spectrum of the test image “keyboard.”
2) (5%) Enforce odd symmetry on the kernel. Show the kernel.
3) (5%) Show the result of frequency-domain filtering of the test image using the vertical Sobel kernel.
4) (5%) Compare your result in 3) with the result of space-domain filtering.
5) (5%) Show the result of frequency-domain filtering without enforcing odd symmetry on the kernel.

keyboard.tif

HW 3(B) Problem Statement

This image capturing one of the circumpolar cyclones on Jupiter's north pole was taken by the JunoCam imager aboard NASA's Juno. The horizontal lines and overall graininess of the image show the effects of radiation damage on the camera. In this homework, you are asked to repair this image and enhance the details by spatial/frequency-domain filtering and intensity transformation.

1) (15%) Remove the horizontal noise by applying frequency-domain filtering. Show the designed filter and the filtered image.
2) (20%) Enhance the surface details and reduce graininess of the filtered image obtained in Part 1 by techniques described in the lecture notes and Fig. 3.57. Show the sharpened image and explain your choice of the techniques.
3) (20%) Enhance the contrast of the sharpened image obtained in Part 2 by histogram equalization. Show the equalized image and the histograms before and after equalization.
4) (20%) Design your own image enhancement procedure for the noisy image. You may use any technique beyond those described in the textbook. Describe the procedure of your technique by a flowchart.



noisy_image.tif

Software Package Allowed

- Python 3.8+
- Standard Python library
- Numpy
- Opencv-python

Assignment Requirements

- Only the Image IO of the Software Package is allowed, other image operations (kernel, filtering, etc.) should be written by yourself.
- Set your directory structure as follows:

  hw03_<your ID>/

  - program files (part_A.py, part_B.ipynb,…)
  - report.pdf
  - images/
    - Images of program output

Assignment Submission Requirements

- Submit to NTU COOL
- Do NOT copy homework (code, report, results, etc.) from others